Antiderivatives and Riemann Sums

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic antiderivatives**

Find the general antiderivative of each function:

a $f(x) = x^4$ $= \frac{x^5}{5} + C$

b $f(x) = \frac{1}{x}$ $= \ln|x| + C$

c $f(x) = e^x$ $= e^x + C$

d $f(x) = \sec^2 x$ $= \tan x + C$

(a) $= \frac{x^5}{5} + C$; (b) $= \ln|x| + C$; (c) $= e^x + C$; (d) $= \tan x + C$

2 **[Foundational]** • **Target: Basic antiderivatives**

Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.

$F(x) = 2x^3 - 4x + C$. $F(1) = 2 - 4 + C = 5 \Rightarrow C = 7$. $F(x) = 2x^3 - 4x + 7$.

3 **[Foundational]** • **Target: Riemann sums: left, right, and midpoint**

Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; left endpoints $x = 0, 1, 2, 3$: $L_4 = 1[0^2 + 1^2 + 2^2 + 3^2] = 1[0 + 1 + 4 + 9] = 14$.

4 **[Foundational]** • **Target: Riemann sums: left, right, and midpoint**

Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; right endpoints $x = 1, 2, 3, 4$: $R_4 = 1[1 + 4 + 9 + 16] = 30$.

5 **[Foundational]** • **Target: Riemann sums: left, right, and midpoint**

Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; midpoints $x = 0.5, 1.5, 2.5, 3.5$: $M_4 = 1[0.25 + 2.25 + 6.25 + 12.25] = 21$. (True value: $\frac{64}{3} \approx 21.33$.)

6 **[Foundational]** • **Target: The Trapezoidal Rule**

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.

$T_4 = \frac{L_4 + R_4}{2} = \frac{14 + 30}{2} = 22$.

7 **[Foundational]** • **Target: The Trapezoidal Rule**

A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.

With 3 equal subintervals of width 2: $T = 2\left[\frac{f(0)+f(2)}{2} + \frac{f(2)+f(4)}{2} + \frac{f(4)+f(6)}{2}\right] = 2\left[\frac{7}{2} + \frac{14}{2} + \frac{17}{2}\right] = 7 + 14 + 17 = 38$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Sums as limits: the definite integral**

Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$ and x_i^* is any sample point in the i-th subinterval — as $n \to \infty$ (subintervals shrink to zero width), all Riemann sum types (left, right, midpoint) converge to the same value, the exact signed area under f.

9 **[Intermediate]** • **Target: Sums as limits: the definite integral**

Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

Since $f(x) = x^2$ is increasing on $[0, 4]$, the left endpoint of each subinterval gives the smallest function value on that subinterval, so the Left Riemann Sum UNDERESTIMATES the true area (confirmed: $L_4 = 14 < \frac{64}{3} \approx 21.33$).

10 **[Intermediate]** • **Target: Riemann sums from a table of values**

A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 8 + 10 + 9 = 32$.

11 **[Intermediate]** • **Target: Visualizing a Riemann sum approximation**

The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.

Since $f(x) = x^2$ is increasing on $[0, 4]$, right-endpoint rectangles reach up to the curve's height at the RIGHT edge of each strip, so their tops poke above the curve, overestimating the area. Left-endpoint rectangles use the height at the LEFT edge (lower, since f is increasing), so their tops stay below the curve, underestimating the area.

12 **[Intermediate]** • **Target: An applied antiderivative problem**

A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

$P(t) = \int r(t)\,dt = 100t + 10t^2 + C$. Using $P(0) = 0$: $C = 0$. $P(t) = 100t + 10t^2$.
$P(10) = 1000 + 1000 = 2000$ units.

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13 **[Intermediate]** • **Target: More antiderivative practice**

Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

$= x^4 + \frac{2}{x} + \sin x + C$.

14 **[Intermediate]** • **Target: Comparing Riemann sums to the exact integral**

For $\int_0^4 x^2\,dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.

Distances from 21.33: M_4 ($|21 - 21.33| = 0.33$), T_4 ($|22 - 21.33| = 0.67$), L_4 ($|14 - 21.33| = 7.33$), R_4 ($|30 - 21.33| = 8.67$). Order (closest to farthest): M_4, T_4, L_4, R_4. The midpoint approximation tends to be more accurate than the trapezoidal for smooth curves, because its errors are typically about half the size and opposite in sign to the trapezoidal error.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Riemann sums from a table of values** • **Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

Identify the first term and ratio: When $n = 1$, $a = 3\left(-\frac{2}{5}\right)^1 = -\frac{6}{5}$. The common ratio is $r = -\frac{2}{5}$. Since $|r| = \frac{2}{5} < 1$, the series converges. Using the sum formula $S = \frac{a}{1-r}$: $S = \frac{-6/5}{1-(-2/5)} = \frac{-6/5}{7/5} = -\frac{6}{7}$

16 **[Challenging]** • **Target: Antiderivatives of trigonometric and exponential functions** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0, 1]$ and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$.

Factor $\frac{1}{n} = \Delta x$: $\sum_{i=1}^{n} \frac{i^3}{n^4} = \sum_{}^{n} \left(\frac{i}{n}\right)^3 \cdot \frac{1}{n}$. Here $x_i = \frac{i}{n} = 0 + i\Delta x$ with $\Delta x = \frac{1}{n}$, matching the right Riemann sum of $f(x) = x^3$ on $[0, 1]$: $\lim_{n\to\infty} \sum_{i=1}^{} \left(\frac{i}{n}\right)^3 \frac{1}{n} = \int_0^1 x^3\,dx = \left[\frac{x^4}{4}\right]_0^1 = \frac{1^4}{4} - 0 = \mathbf{\frac{1}{4}}$.

17 **[Challenging]** • **Target: Visualizing a Riemann sum approximation** • **Cross-Lesson: 8.01 Area Between Curves**

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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18 [Challenging] • Target: An applied antiderivative problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x-3}{x^2-2x-3}\,dx$

Factor the denominator: $x^2-2x-3=(x-3)(x+1)$. Set up partial fractions: $\frac{5x-3}{(x-3)(x+1)}=\frac{A}{x-3}+\frac{B}{x+1}\Rightarrow 5x-3=A(x+1)+B(x-3)$ Using Heaviside cover-up: Set $x=3$: $5(3)-3=A(3+1)\Rightarrow 12=4A\Rightarrow A=3$. Set $x=-1$: $5(-1)-3=B(-1-3)\Rightarrow -8=-4B\Rightarrow B=2$. Integrate term-by-term.

19 [Challenging] • Target: More antiderivative practice • Cross-Lesson: 1.01 Limits Graphically and Numerically

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Step 1: Test for absolute convergence. Consider $\sum|a_n|=\sum_{n=1}^{\infty}\frac{1}{n^{1/2}}$. This is a p-series with $p=1/2\le 1$, which diverges. Hence the series does NOT converge absolutely. Step 2: Apply Alternating Series Test. Here $b_n=\frac{1}{\sqrt{n}}>0$. 1. Monotonicity: \(n+1 > n \implies \sqrt{n+1} > \sqrt{n} \implies \frac{1}{\sqrt{n+1}} conditionally convergent.

20 [Challenging] • Target: Comparing Riemann sums to the exact integral • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Does $\sum_{n=0}^{\infty}(0.9)^n$ converge, and to what?

Method · Geometric Series Convergence & Sum: Step 1 · Identify series parameters: Initial term: $a=(0.9)^0=1$. Common ratio: $r=0.9=\frac{9}{10}$. Step 2 · Test convergence: $|r|=0.9<1$, therefore the series converges. Step 3 · Calculate exact sum: $S=\frac{a}{1-r}=\frac{1}{1-0.9}=\frac{1}{0.1}=\mathbf{10}$. Conclusion: Converges to **10**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic antiderivatives

Find C in the particular antiderivative $F(x)=x^3-7x+C$ of $f(x)=3x^2-7$, satisfying $F(-1)=-8$.

a. -8 **b.** -12

c. -14 ✓ **d.** -16

$F(-1)=(-1)^3-7(-1)+C=6+C=-8$, so $C=-14$.

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22 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^6 f(x)\,dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 7$, $f(2) = 12$, $f(4) = 9$, $f(6) = 3$.

a. 54

b. 52 ✓

c. 62

d. 50

$T = 2\left[\frac{7+12}{2} + \frac{12+9}{2} + \frac{9+3}{2}\right] = 2\left[\frac{7+3}{2} + 12 + 9\right] = 2(5 + 12 + 9) = 52$.

23 [Foundational] • Target: The Trapezoidal Rule

For $f(x) = 4x^2$ on $[0, 5]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^5 f(x)\,dx$.

a. Both Left and Right Riemann Sums underestimate.

b. Both Left and Right Riemann Sums overestimate.

c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

d. **Left Riemann Sum underestimates, and Right Riemann Sum overestimates.** ✓

$f(x) = 4x^2$ is increasing on $[0, 5]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

24 [Foundational] • Target: Sums as limits: the definite integral

A factory's production rate is $r(t) = 80 + 26t$ units/day, where t is days since a process change. Find the total production $P(2)$ given $P(0) = 0$.

a. 160

b. 238

c. 186

d. 212 ✓

$P(t) = \int r(t)\,dt = 80t + 13t^2 + C$. Using $P(0) = 0$: $C = 0$. $P(2) = 80(2) + 13(2)^2 = 160 + 52 = 212$ units.

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25 [Foundational] • Target: Riemann sums from a table of values

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:

a. $n \to \infty$ (partition width $\to 0$) ✓

b. $b \to \infty$

c. $f \to 0$

d. $x \to a$

The definite integral is defined as the limit of Riemann sums as the number of subintervals $n \to \infty$.

26 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4 ✓

d. $-4i$

Working through this complex-numbers step: $z = 4\cos 180^\circ + 4\sin 180^\circ\, i = -4$

27 [Foundational] • Target: Visualizing a Riemann sum approximation

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

a. $\ln(2)$

b. 1

c. 2 ✓

d. $\ln(1)$

First, we simplify the function $f(x)$. Using the double angle identity for sine, $\sin(2x) = 2\sin(x)\cos(x)$. Therefore, $f(x) = \frac{2\sin(x)\cos(x)}{\cos(x)}$. For $\cos(x) eq 0$, we can cancel the $\cos(x)$ terms, so $f(x) = 2\sin(x)$. Now we need to evaluate the definite integral: $\int_0^{\pi/2} 2\sin(x)dx$. The antiderivative of $2\sin(x)$ is $-2\cos(x)$. Evaluating the definite integral using the Fundamental Theorem of Calculus: $[-2\cos(x)]_0^{\pi/2} = -2\cos(\pi/2) - (-2\cos(0))$.

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28 [Foundational] • Target: An applied antiderivative problem

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

a. $x - y + 2z = -4$

b. $x + y + 2z = 6$

c. $x - y + 2z = 4$ ✓

d. $x + y = -2$

A normal vector is $\vec{PQ} \times \vec{PR}$, a multiple of $\langle 1,\ -1, 2\rangle$. Using the point $(-3,\ 1,\ 4)$ gives $x - y + 2z = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: More antiderivative practice

Find the arc length of $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}\,t\rangle$ for $0 \le t \le \ln(2)$.

a. $\frac{3}{2}$ ✓

b. 3

c. $\frac{3}{4}$

d. $\frac{5}{2}$

$L = \int_a^b ||\mathbf{r}'(t)||\,dt$; here $||\mathbf{r}'(t)||$ simplifies to a function with an elementary antiderivative, and the integral equals $\frac{3}{2}$.

30 [Intermediate] • Target: Comparing Riemann sums to the exact integral

Find the area of the triangle with vertices $(1,\ 0,\ 2)$, $(1,\ 1,\ 2)$, $(-3,\ 3,\ -3)$.

a. $\sqrt{41}$

b. $\frac{\sqrt{41}}{3}$

c. $\frac{\sqrt{41}}{2}$ ✓

d. $\frac{3}{2}$

$\vec{PQ} \times \vec{PR} = \langle -5, 0, 4\rangle$, whose length is $\sqrt{41}$. The area is half of that: $\frac{\sqrt{41}}{2}$.

31 [Intermediate] • Target: Basic antiderivatives

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. **0** ✓

b. ∞

c. 1

d. 6

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{6}{n+1} \to 0$ as $n \to \infty$. Since $L = 0 < 1$, the series converges (for every x).

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32 [Intermediate] • Target: Riemann sums: left, right, and midpoint

Gradient vector $ablaf = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$. For $f = x^2 + y^2$ at $(1, 1, 0)$:

a. $(1, 1, 0)$

b. $(1, 2, 0)$

c. $(0, 0, 0)$

d. $(2, 2, 0)$ ✓

Applying Vectors and Vector-Valued Functions principles, $ablaf = (2x, 2y, 0) = (2, 2, 0)$. Therefore, the verified correct option is $(2, 2, 0)$.

33 [Intermediate] • Target: The Trapezoidal Rule

Continuity of piecewise: $f(x) = \{kx, x < 2; x^2 - 2, x \geq 2$ is continuous iff $k =$

a. 1 ✓

b. 2

c. 0

d. 3

Applying Limits and Continuity principles, $2k = 4 - 2 = 2 \Rightarrow k = 1$. Therefore, the verified correct option is 1.

34 [Intermediate] • Target: Sums as limits: the definite integral

Let $f(x) = x^3 + 3x - 6$, which is one-to-one. Find $(f^{-1})'(-6)$.

a. $\frac{1}{3}$ ✓

b. 3

c. $\frac{1}{111}$

d. $-\frac{1}{6}$

$f(0) = -6$, so $(f^{-1})'(-6) = \frac{1}{f'(0)} = \frac{1}{3}$ since $f' = 3x^2 + 3$.

35 [Intermediate] • Target: Riemann sums from a table of values

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$.

a. 0

b. 2

c. ∞ ✓

d. $\frac{1}{2}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2|x|}{n+1} \to 0$ for every x, so $R = \infty$.

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36 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find the distance from $(2,\ -2,\ 0)$ to the plane $-3y+z=-4$.

a. 10

b. $\frac{\sqrt{10}}{5}$

c. 1

d. $\sqrt{10}$ ✓

$D=\frac{|ax_0+by_0+cz_0-d|}{\sqrt{a^2+b^2+c^2}}=\frac{10}{\sqrt{10}}=\sqrt{10}.$

37 [Intermediate] • Target: Visualizing a Riemann sum approximation

For the Dirichlet function $f(x)=\{1, x\in\mathbb{Q};\ 0, x \text{otin} \mathbb{Q}$, $\lim_{x\to 0} f(x)$:

a. equals 0

b. equals 1

c. equals $\frac{1}{2}$

d. **does not exist** ✓

In every interval around 0, f takes both values 0 and 1, so no limit exists.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: An applied antiderivative problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Evaluate $\tan^{-1}(1)$ in degrees.

a. 67°

b. 22°

c. 69°

d. 45° ✓

Working through this functions step: $\tan^{-1}(1)=45^\circ$

39 [Challenging] • Target: More antiderivative practice • Cross-Lesson: 1.01 Limits Graphically and Numerically

Telescoping series $\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)=$

a. ∞

b. 1 ✓

c. 0

d. 1/2

Partial sums: $S_n=1-1/(n+1)\to 1$.

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40 [Challenging] • Target: Comparing Riemann sums to the exact integral • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞ ✓

c. π

d. 2

Ratio test: $L = 0 \Rightarrow R = \infty$.

41 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 2.01 The Definition of the Derivative

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. **True** ✓

c. Only for separable functions

d. Depends on Fubini

Inner integral is evaluated first.

42 [Challenging] • Target: Riemann sums: left, right, and midpoint • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1, 5)$ ✓

b. $[-1, 5]$

c. $(-1, 5]$

d. $[-1, 5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R = 3$ and $a = 2$. Thus, the series converges for $|x-2| < 3$. This inequality can be rewritten as $-3 < x-2 < 3$. Adding 2 to all parts of the inequality, we get $-3+2 < x < 3+2$, which simplifies to $-1 < x < 5$. This interval is written in open interval notation as $(-1, 5)$. The convergence at the endpoints.

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43 [Challenging] • Target: The Trapezoidal Rule • Cross-Lesson: 3.01 The Chain Rule

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

44 [Challenging] • Target: Sums as limits: the definite integral • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?

a. $-\frac{4}{3}$ ✓

b. 0

c. $\frac{4}{3}$

d. 2

To find the definite integral of $f(x) = x^2 - 4x + 3$ from $x = 1$ to $x = 3$, we evaluate $\int_1^3 (x^2 - 4x + 3)\,dx$. First, find the antiderivative of $f(x)$: $F(x) = \frac{x^3}{3} - \frac{4x^2}{2} + 3x = \frac{x^3}{3} - 2x^2 + 3x$. Next, apply the Fundamental Theorem of Calculus: $F(3) - F(1)$. Calculate $F(3)$: $F(3) = \frac{3^3}{3} - 2(3)^2 + 3(3) = \frac{27}{3} - 2(9) + 9 = 9 - 18 + 9 = 0$. Calculate $F(1)$: $F(1) = \frac{1^3}{3} - 2(1)^2 + 3(1) = \frac{1}{3} - 2 + 3 = \frac{1}{3} + 1 = \frac{4}{3}$. Fi.

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45 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \leq x < 2$

b. $-1 < x \leq 2$ ✓

c. $-1 \leq x \leq 2$

d. $-1 < x < 2$

Let the given power series be $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{(2x-1)^n}{n \cdot 3^n}$. We will use the Ratio Test to find the interval of convergence. $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \to \infty} \left| \frac{(2x-1)^{n+1}}{(n+1) \cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{(2x-1)^n} \right|$.

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